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SECTION 02

LECTURER: DR. ROLIANA BINTI IBRAHIM

NAME	MATRIC NUMBER
NURUL IKA SYAFINY BINTI AZHAR	A23CS0164
LUBNA AL HAANI BINTI RADZUAN	A23CS0107
NURAI SYAH BINTI MOHD ZIKRE	A23CS0160
NUR FIRZANA BINTI BADRUS HISHAM	A23CS0156
HARINI A/P SANGARAN	A23CS0081

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Phase 1

1.0 Introduction

MindConnect is an innovative Artificial Intelligence (AI)-Based Diagnostic Support System designed for the preliminary detection and safe management of mental health conditions. The system employs a hybrid knowledge representation architecture to manage the complexity of psychological data, ensuring personalized and safe interventions.

The core functions rely on specific computational models:

- **Rule-Based Systems** : Used for preliminary diagnosis rules (2.1).
- **Frame-Based Systems** : Used for therapy selection by structuring user progress and severity data (2.2).
- **Semantic Networks** : Used for modeling user relationships (emotion, trigger, coping strategy) and risk assessment (2.3).
- **Predicate Logic** : Used for crisis triage by handling the uncertainty and degrees of truth in critical inputs (2.4).
- **Linguistic Patterns** : Used for language analysis to translate text into actionable facts about mood and issue dwelling (2.5).

Operating under a strict Ethical Rule against formal clinical diagnosis, MindConnect focuses solely on early detection, personalized support, and safe hand-offs.

2.0 Knowledge Representation

Knowledge Representation (KR) in AI refers to transforming complex, real-world knowledge like facts, rules, and relationships into a structured, formal language such as First-Order Logic (FOL) or network models that the AI can process logically. The goal of KR is to enable inference, allowing the system to derive new conclusions from existing information.

MindConnect employs KR schemes as the backbone of its reasoning capabilities. Specifically, the system uses a hybrid approach to handle the distinct types of psychological data it encounters:

- **Rule-Based Systems** define the "**what-if**" **rules** for diagnosis and crisis triage.
- **Frame-Based Systems** create the structured, digital **user profile** needed for personalized therapy selection.
- **Semantic Networks** map the **relationships** between concepts or emotions for contextual understanding and risk assessment.

By structuring raw input from Natural Language Processing (NLP) into these formal models, MindConnect ensures its decisions, whether a preliminary diagnosis, a change in therapy, or a safety check-in are consistent, logical, and safe.

2.1 First Order Logic (FOL) in MindConnect's for Preliminary Diagnosis

A Rule-Based System (RBS) is a knowledge representation scheme that uses a collection of IF-THEN statements to encode knowledge and perform logical reasoning. It is a foundational technology in artificial intelligence that has long been used in decision-making and problem-solving across various domains (GeeksforGeeks, 2025). MindConnect utilizes a RBS as the foundation for its AI-Based Diagnostic System. Its primary role is to ensure early detection of mental health conditions like Generalized Anxiety Disorder (GAD) and Mild-to-Moderate Depression. Its rules map user-reported symptoms extracted via NLP from the conversation to potential conditions. Crucially, the RBS is constrained by the Ethical Rule that the AI will never attempt to give a formal clinical diagnosis. For safety constraints, it is designed only for support, detection, and safe hand-offs. Below is First-Order-Logic (FOL) for GAD and Mild Depression diagnosis.

Rule 1: Indicating Generalized Anxiety Disorder (GAD)

This rule formalizes the logic for indicating GAD based on persistent, excessive worry and an associated physical symptom (Mayo Clinic, n.d.).

$$\forall P (Symptom(P, ExcessiveWorry) \wedge Symptom(P, Restlessness) \wedge ReportsDuration(P, Weeks)) \rightarrow Indicates(P, GAD)$$

Natural Language: For all persons (P), if P reports Excessive Worry AND this has lasted Weeks AND P reports Restlessness, THEN it indicates a potential Generalized Anxiety Disorder (GAD).

Rule 2: Indicating Mild-to-Moderate Depression

This rule formalizes a key diagnostic criterion for depression: the presence of a persistent low mood combined with a loss of pleasure (anhedonia) (Silva S., 2024).

$$\forall P (Symptom(P, LowMood) \wedge Symptom(P, Anhedonia) \wedge ReportsDuration(P, Weeks)) \rightarrow Indicates(P, MildDepression)$$

Natural Language: For all persons (P), if P reports Low Mood AND reports Anhedonia (loss of interest or pleasure) AND this has lasted Weeks, THEN it indicates a potential Mild Depression.

2.2 First-Order Logic (FOL) in MindConnect's for Therapy Selection

Frame-Based Systems can transform raw conversational input into structured knowledge that is essential for the logical reasoning engine by structuring a complex psychological information into discrete slots (e.g., Progress, Severity). Frame-Based Systems provide the structural architecture for this organization where it serves as the digital profile for each user, modeling their current and historical mental state (Minsky, 1974). By). This ensures the AI can quickly and consistently retrieve all necessary data points to select the safest and most effective therapeutic response.

Besides, a frame is ideally suited to model a single user's mental profile or a specific therapeutic intervention, hence, frame-based systems are critical to MindConnect because they bridge the gap between unstructured conversational data (what the user types) and structured, actionable knowledge required for intelligent decision-making (Minsky, 1974). This structure allows the MindConnect to efficiently achieve its goals of personalization, immediacy, and safety.

Rule 1: Intervention Selection Based on Stagnant Progress

This rule dictates a mandatory escalation of therapy when the user's current intervention is failing, focusing on the goal of continuous improvement.

$$\forall P : (HasProfile(P, Anxiety) \wedge Progress(P, Stagnant) \wedge (Severity(P) > 8)) \rightarrow Needs(P, NewExercise)$$

If any user (P) has an anxiety profile *AND* their progress is stagnant *AND* their severity score is above the threshold (8), THEN they need an elevated intervention.

Rule 2: Low Severity Intervention Default

This rule defines the conditions under which the system provides supportive, non-intensive check-ins, ensuring resources are appropriately managed for users who are stable or improving.

$$\forall P : (HasProfile(P, MildDepression) \wedge Progress(P, Improving) \vee (Severity(P) \leq 5)) \rightarrow Needs(P, DailyCheckin)$$

For any user (P), if they have a mild depression profile *AND* their progress is improving *OR* their severity score is low (≤ 5), THEN they only require the standard, supportive *DailyCheckIn* to maintain stability and monitor their state.

2.3 First-Order Logic (FOL) in MindConnect's for User Relationships

Semantic Networks enable MindConnect to organize psychological knowledge as interconnected nodes and relationships, allowing the system to represent concepts such as *User*, *Emotion*, *Trigger*, and *Coping Strategy* in a structured and relational manner. Instead of storing information as isolated data points, semantic networks map how these concepts influence or depend on one another—for example, *Stress* $\rightarrow$ *caused_by* $\rightarrow$ *Workload* or *Anxiety* $\rightarrow$ *alleviated_by* $\rightarrow$ *BreathingExercise*. This relational mapping helps the AI interpret user input within a broader psychological context, enabling it to recognize how multiple factors contribute to a user's mental state.

By modeling mental-health concepts as a network rather than a list, MindConnect can perform more advanced reasoning and inference. The system can trace emotional patterns, identify relevant interventions, and anticipate potential risks based on connected nodes. These structured relationships help the AI generate responses that are not only accurate but also contextually

sensitive and consistent with established psychological principles. In doing so, semantic networks play a crucial role in supporting MindConnect’s goals of delivering personalized, safe, and empathetic digital mental-health support.

Rule 1: Identifying Emotional Risk Through Negative Sentiment

This rule ensures the AI detects high-risk emotional states early to support safety and timely intervention.

Semantic Network Relationship

User $\rightarrow$ *HasEmotion* $\rightarrow$ NegativeSentiment $\rightarrow$ *HasIntensity* $\rightarrow$ HighIntensity $\rightarrow$ AtRisk

FOL Representation

$\forall U, E \text{ (IsUser}(U) \wedge \text{HasEmotion}(U, E) \wedge \text{Negative}(E) \wedge \text{Intensity}(E, L) \wedge \text{High}(L) \rightarrow \text{AssignRisk}(U, \text{AtRisk}))$

If a user consistently expresses negative emotions *AND* their emotional intensity is high, *THEN* the system must classify the user as “At-Risk” and trigger supportive actions.

Rule 2: Recommending Check-Ins When User Avoids Engagement

This rule maintains continuity in therapy by responding to patterns of disengagement.

Semantic Network Relationship

User $\rightarrow$ *HasEngagementLevel* $\rightarrow$ LowEngagement $\rightarrow$ *RelatedTo* $\rightarrow$ Avoidance $\rightarrow$ *TriggersAction* $\rightarrow$ CheckIn

FOL Representation

$\forall U \text{ ((IsUser}(U) \wedge \text{EngagementLevel}(U, L) \wedge \text{Low}(L)) \vee \text{AvoidsResponse}(U) \rightarrow \text{TriggerCheckIn}(U))$

If a user has been inactive for an extended period *OR* avoids responding to reflective questions, *THEN* the system initiates a wellness check-in to maintain therapeutic connection.

2.4 First-Order Logic (FOL) in MindConnect's for Crisis Triage

The core job of the MindConnect system is to translate the user's various inputs (such as manual ratings, physiological data, or system metadata) into fuzzy inputs that the AI's rules can use. When processing these inputs, the system instantly analyzes them to determine degrees of mood, the severity of reported symptoms, and records the time of day. By converting these raw, imprecise measurements into degrees of membership in fuzzy sets, the system gives its fuzzy inference rules the input to work.

To achieve the goals of preliminary diagnosis and early detection, the MindConnect requires a strong way to represent human emotional and psychological states in a nuanced manner, a key approach when applying computational models to subjective areas like psychology (Deshpande, 2019). The system relies on its fuzzy rules to transform numerical inputs into actionable inferences for personalized responses and appropriate crisis triage that capture degrees of uncertainty, which is particularly useful in clinical decision support systems (Warren et al., 2000).

Rule 1: Identifying Emotional Distress

This rule triggers a responsive suggestion when a significant degree of negative self-rating and a high frequency of physical stress indicators are detected, suggesting a need for validation and structured therapeutic interaction.

Rule Syntax:

IF (User_Rating IS Poor) AND (Heart_Rate IS Elevated) THEN (Anxiety_Level IS High)

- The User_Rating input (e.g., a numerical value from 1 to 10) has a specific degree of membership in the fuzzy set Poor (e.g., $\mu=0.8$).
- The Heart_Rate input (e.g., bpm) has a specific degree of membership in the fuzzy set Elevated (e.g., $\mu=0.9$).
- The rule is evaluated using the minimum operator ($\min(0.8, 0.9) = 0.8$), resulting in a firing strength that contributes to the final Anxiety_Level output.

Rule 2: Triggering Immediate Coping Action

This rule is designed for immediacy and crisis prevention, linking environmental context with emotional tone (e.g., anxiety) to deploy a specific and tailored coping mechanism.

Rule Syntax:

IF (Time_of_Day IS Night) AND (Anxiety_Level IS High) THEN (Offer_Intervention IS BreathingGuide)

- The Time_of_Day input (e.g., an hour value) has a specific degree of membership in the fuzzy set Night.
- The Anxiety_Level has a specific degree of membership in the fuzzy set High.
- The system uses the degrees of truth for both conditions to infer a weighted recommendation for a guided coping mechanism like a 2-minute breathing guide.

Rule 3: Crisis Triage

This rule handles crisis triage by escalating the response when indicators cross a critical threshold, requiring a high-level intervention.

Rule Syntax:

IF (Self_Harm_Indicator IS Present) OR (Suicide_Ideation IS Extreme) THEN (Triage_Level IS Emergency)

- The Self_Harm_Indicator has a high degree of membership in the fuzzy set Present based on specific inputs.
- The Suicide_Ideation input has a high degree of membership in the fuzzy set Extreme. The system uses the maximum operator for the OR condition ($\max(\mu_{Present}, \mu_{Extreme})$).
- If the resulting degree of truth is high enough, the Triage_Level output is strongly associated with Emergency, which triggers an immediate alert to human crisis support or provides a list of emergency resources.

2.5 First-Order Logic (FOL) in MindConnect's for Language Analysis

The core job of Natural Language Processing (NLP) in MindConnect is to translate the user's feelings and thoughts into facts that the AI's rules can use. When sending the message, NLP instantly analyzed the text to determine the mood. This is to check if the user is stuck in thinking and records the time of day. By converting the complex language into simple facts, NLP gives the AI's logic rules the input to work.

To achieve the goals of preliminary diagnosis and early detection, the MindConnect requires a strong way to represent human emotional and psychological states. Given the reliance on Natural Language Processing (NLP) for analyzing user input (Glaz et al., 2020), these rules transform structured facts into actionable inferences for personalized responses.

Rule 1: Identifying Emotional Distress

This rule triggers an immediate, reflective response when negative language and repetitive dwelling are detected, suggesting a need for validation and structured therapeutic interaction.

$$\forall T (Mood(T, Down) \wedge StuckOnIssue(T) \rightarrow NeedsSupport(T))$$

The user's text in turn T expresses sadness, worry or any general unhappiness. The user's language pattern in turn T shows them endlessly repeating the same problem. The AI infers that the user is currently in emotional distress and requires an immediate therapeutic response.

Rule 2: Triggering Immediate Coping Action

This rule is designed for immediacy and crisis prevention, linking environmental context with emotional tone (e.g. anxiety) to deploy a specific and tailored coping mechanism (Horstkötter, 2025).

$$\forall U, E (Time(E, Night) \wedge Feels(E, Nervous) \rightarrow Offer(U, BreathingGuide))$$

The user's input event E occurred during non-standard hours (e.g. 10 pm to 6 am). The user's language tone in event E is classified as anxious, stressed or worried. The AI should immediately prompt the user U with a quick and guided coping mechanism like a 2-minutes breathing guide.

Phase 2

1.0 State Space Search Diagram

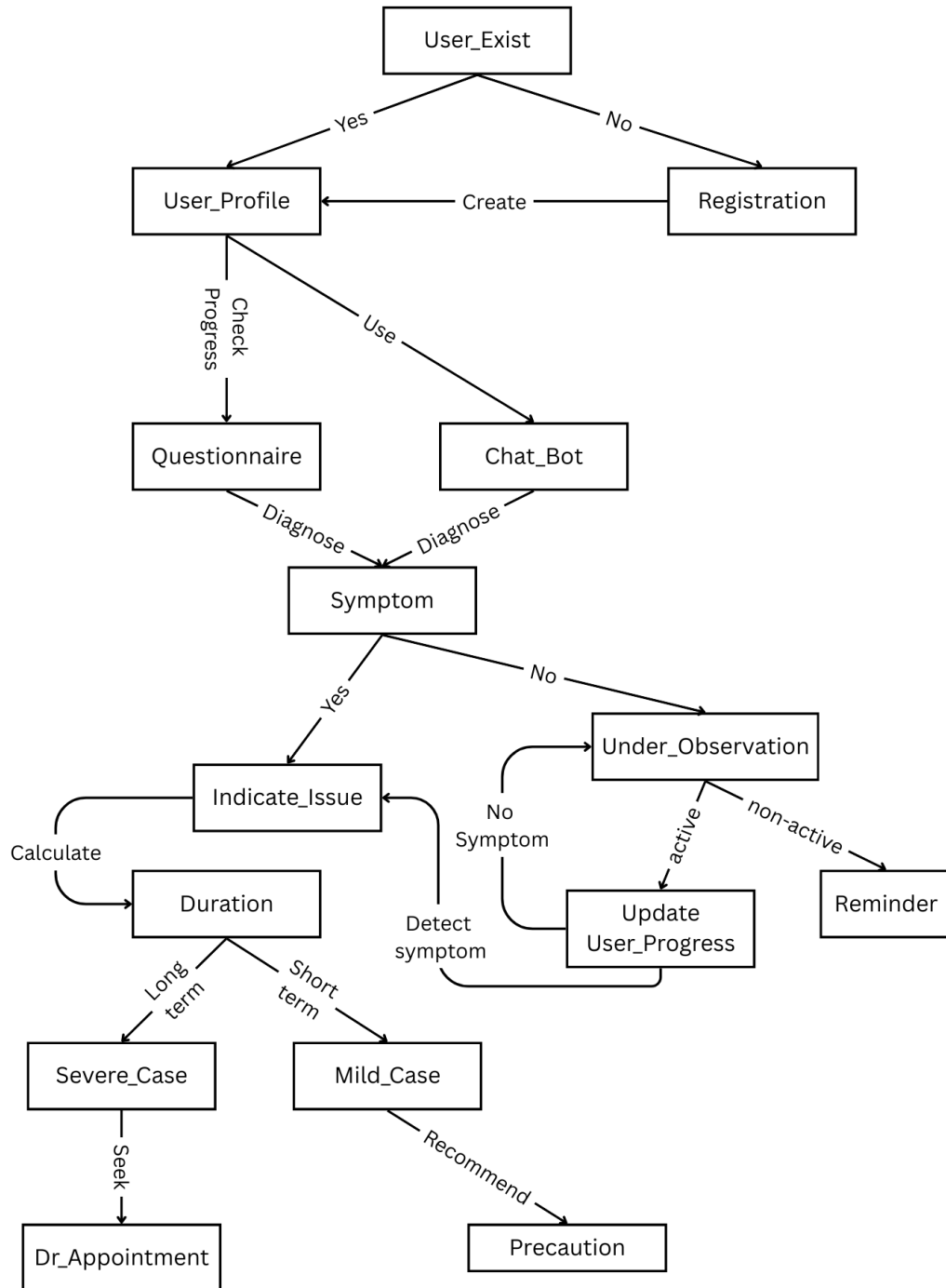


Figure 1: MindConnect State Space Search Diagram

Initial states: User_Exist

Actions: User_Profile, Registration, Questionnaire, Chat_Bot, Symptom, Indicate_Issue, Under_Observation, Duration, Update User_Progress, Severe_Case, Mild_Case

Goals: Dr_Appointment, Precaution, Reminder

Path cost: Yes, No, Create, Check Progress, Use, Diagnose, Active, Non-active, No Symptom, Detect symptom, Calculate, Long term, Short term, Seek, Recommend

Explanation:

The process begins at the Initial State by determining whether a User_Exist. If the user is new, the Registration action will be triggered by the system to Create a User_Profile. The profile of the user will be the central repository for all process and historical progress.

Once the user has activated their profile, the system will move into a diagnostic state where the user can choose to take a Questionnaire or use a Chat_Bot interface. These two interfaces utilize Natural Language Processing (NLP) and Linguistic Patterns to Diagnose users based on their symptoms shown when they are using the interface. From the Symptom obtained (Yes) from the diagnosis, the system will transition to the Indicate_Issue state. In this stage, the systems will calculate the Duration for evaluation to determine whether it has lasted for long term or short term in order to determine the severity of the case. This transition represents the execution of First-Order-Logic (FOL) as it is used to indicate a specific condition of a disease.

Different severity and duration will lead to different results, where the long term and Severe_Case user will need to Seek for Dr_Appointment. As for users with short term and Mild_Case, the system will Recommend a Precaution for an early precaution. Hence, this search has reached its Goal State.

If no Symptom detected (No), users will enter the Under_Observation state, where this state will utilize Semantic Networks to monitor the engagement of the users. If the user remains active, the system will Update User_Progress, where in this state the system will periodically loop back to Symptom state to update the user's emotional state. Next, if the user is non-active, the system will issue a Reminder to update their emotional state.

Phase 3

1.0 Intelligent Agent

The intelligent agent used in MindConnect is a learning agent. The agent takes keystrokes as sensory input and acts on those inputs and displays output on the screen providing solutions to the users' problem.

2.0 PEAS Model Representation

This project is utilizing a learning agent type to control the system. Learning agents work to receive user input to understand users. Initially creating a psychological baseline for a user using a structured questionnaire. Subsequently, the agent will update the user's psychological condition using real-time engagement which is an AI-powered chatbot with the user to detect any sign of emotional trigger and determine the current state of the user's psychology. From there, the agent will produce an output as response based on the user's condition categorized from mild to critical to help the user in-need.

Table 1: PEAS Table Description

Performance	1. Accuracy of psychological trigger and emotion detection 2. Accuracy of psychological responses to the user
Environments	1. User reaction and response : User's emotional state and their subsequent behavior after receiving a reply
Actuators	1. Trigger Alert: Notifying the user of a detected pattern or reminder for the user's daily or weekly update. 2. Psychological Precautions: Delivering guides, reminders, or calming techniques to reduce the trigger impact. 3. Medical Assistance: Facilitating the user to connect with psychological experts.
Sensors	1. Keyboard input: User's response from questionnaire and chatbot. 2. NLP Analyzer: Understanding user's behaviour condition and pattern

	recognition. 3. System Clock: The time history when triggers commonly happen and its frequency.
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3.0 Diagram Representation

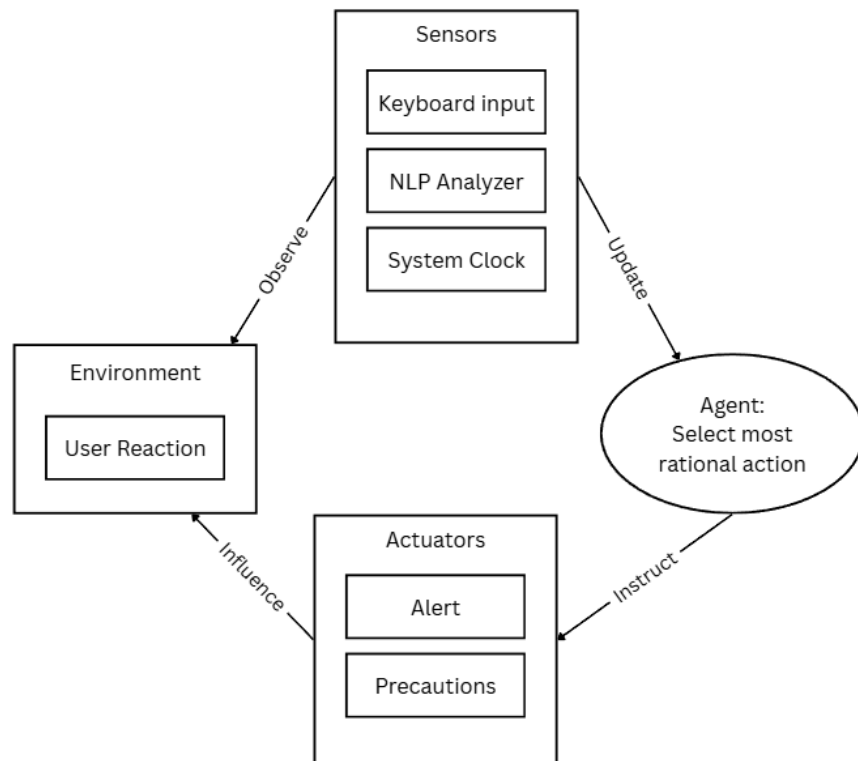


Figure 2: PEAS Diagram Representation

MindConnect is an AI application that detects and responds to user emotions based on the interaction in the application. Its performance is measured by how accurately it can identify the user's emotional state from the input provided. Higher accuracy ensures the agent responds appropriately and effectively.

The process begins with MindConnect gathering information about users' emotional state via questionnaire and chatbot. Thus, the environment will be a user reaction as it is being monitored.

Keyboard input, Natural Language Processor (NLP) analyzer and system clock are the sensors of this application as it is the source of information from the users itself. Once MindConnect has gained information via questionnaire and chatbot, the application categorises the severity of the condition and updates the agent. The agent of MindConnect then suggests appropriate ways of measure to be taken by users. For example, if the user's condition is mild then MindConnect provides suggestions, therapy and support via chatbot and if the condition is severe, then it suggests users to seek professional medical help. The suggestions and response given by the AI is considered as the actuators. After the conversation has ended with the users, MindConnect still monitors users' condition via daily check-ins and sending notifications if users are not responsive or not consistent, ensuring improving users' state of mind to the better.

Proof of Concept

The core objective of this Proof of Concept (PoC) is to demonstrate how a mobile interface can serve as a screening tool. By utilizing AI-driven analysis of user-reported data the app identifies early warning signs of mental health struggles before they worsen.

1.0 Login and Homepage

Figure 3 shows the login interface of MindConnect app. Users are required to enter their username and password to login to their account. If the users' forgot their password, they can reset their password by clicking on the "Forgot Password?" hyperlink. The app will redirect to the password changing interface as per Figure 4. The users are required to enter the email address associated with their MindConnect app. A reset link will then be sent to the email. Users can reset their password via the link. If the users do not own a MindConnect account, then they can click on the "Sign Up" hyperlink to create an account. Figure 5 shows the interface of the new MindConnect account creation. Users are required to enter their first name, last name, email address and password to create the new account. Once users have logged into the account, they will view the homepage interface as per Figure 6. The homepage displays all the features of the app, which are chatbox, questionnaire, daily mood tracker and profile update feature.

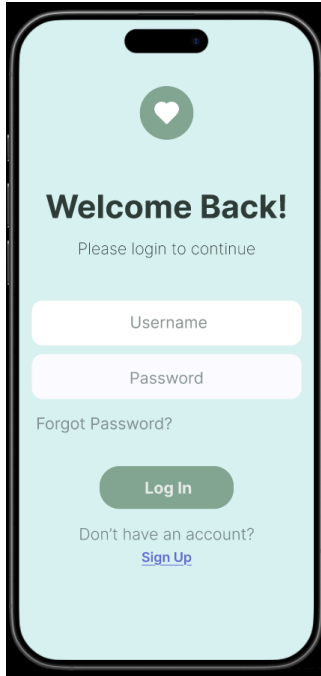


Figure 3 : Login Page

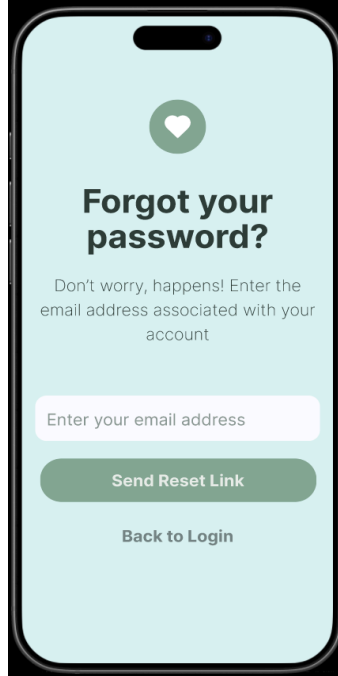


Figure 4 : Password Changing Interface

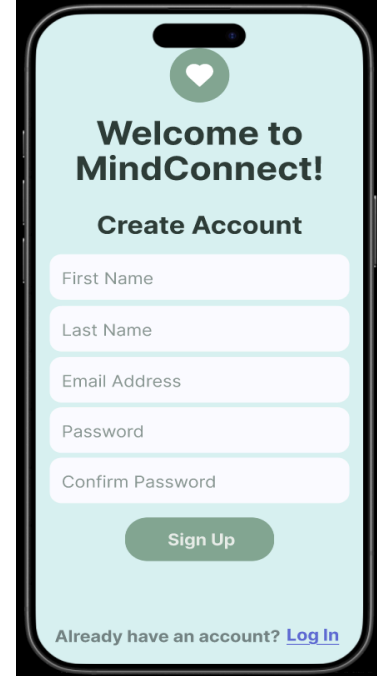


Figure 5 : New MindConnect Account Creation

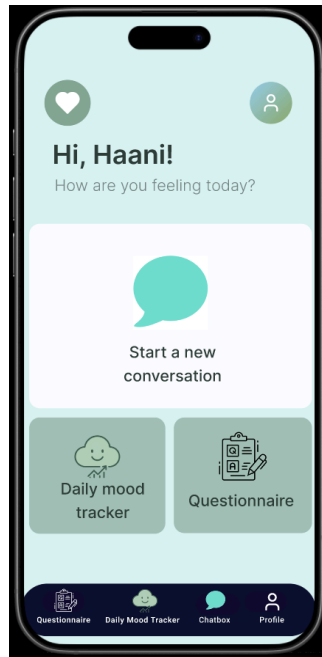


Figure 6 : MindConnect Homepage Interface

2.0 AI Chatbox

This feature allows users to interact with a built-in AI bot to vent their feelings without being embarrassed or feel like they are being judged since they are only talking with a robot. The input and data is gathered from the conversation to diagnose the user's possible problems and its condition. Users can ask anything to the chat bot and they will give useful answers, precautions and words of encouragement based on the conversation. When users want to refer back to their past conversations with the chat bot, they can chat back with the bot regarding the topic or show the conversation to their psychiatrist when needed for treatment since all conversation is saved. These features can be seen in Figure 7 where screenshots of the prototype are captured for better understanding of the AI chat bot function.

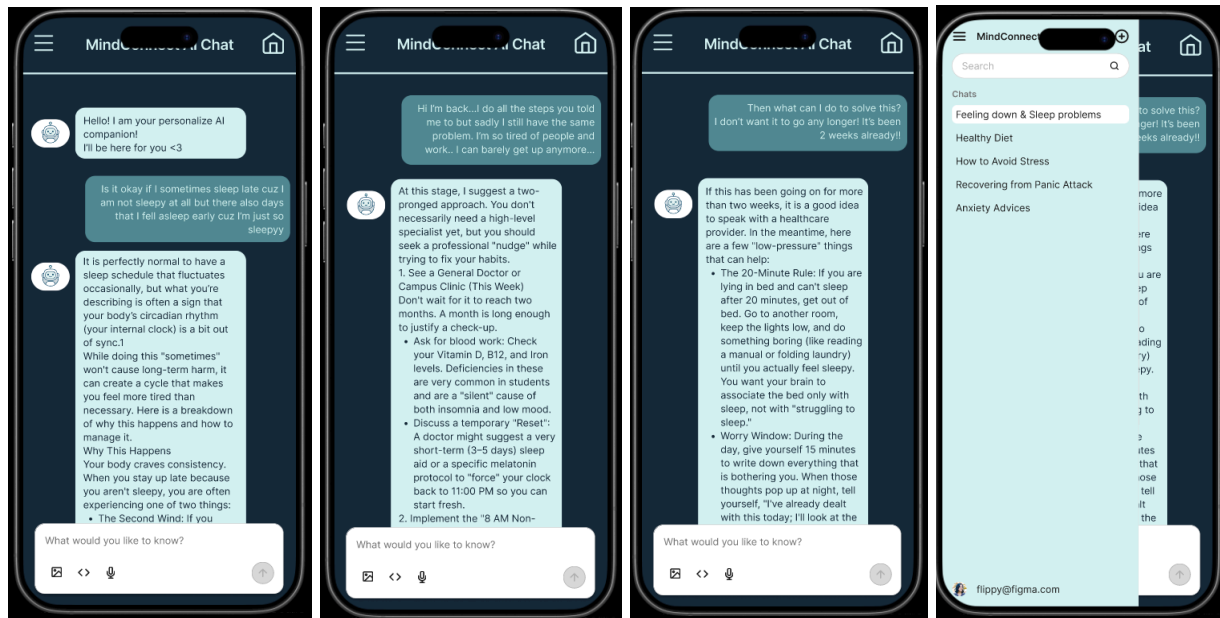


Figure 7: AI Chat Bot Function in MindConnect

3.0 Questionnaire

The questionnaire will act as the main point of data input and it will use quantitative sliders to give the AI numerical information as opposed to the simple binary answers. The duration of the symptoms is monitored by the system and an important measure of mental health diagnosis. This early detection questions as shown in Figure 8.

Have you been experiencing excessive worry or fear?
Feeling anxious about various aspects of life

Not at all 0 Very much

Move the slider to indicate how much this applies to you over the past 2 weeks

Have you felt restless or on edge?
Difficulty relaxing or feeling tense

Calm 0 Very restless

Move the slider to indicate how much this applies to you over the past 2 weeks

How long have you been experiencing these symptoms?
This helps us understand the persistence of your symptom

Less than 2 weeks
Recent onset of symptoms

Less than 2 weeks
Recent onset of symptoms

Figure 8: Questionnaire

After all the 6 questions and duration are answered, the application will process the cumulative of the symptoms. The PoC categorizes the results into three risk tiers as shown in Figure 9 which is under observation, mild symptom detected and significant symptom detected.

Emergency

You're Looking Steady!
Your assessment indicates you're in a stable mental wellness state

Status: Under Observation

Emergency

We're Here to Support You
Your assessment indicates you're in a stable mental wellness state

Status: Mild Symptoms Detected

Emergency

We're Here to Support You
Your assessment indicates you're in a stable mental wellness state

Status: Significant Symptoms Detected

Figure 9: Result after doing the questionnaire

To uphold ethical standards of AI, the PoC will have a disclaimer to explain that the tool is not intended to provide clinical diagnosis but an early-detection tool. A risk-tiering model informs the system to assess the users based on the severity and duration of symptoms as evidenced in Table 2.

Table 2: Risk Classification and AI Intervention Logic

Status	Interpretation	AI-Driven Action
Under Observation	Symptoms are low.	Provides affirmations and encourages healthy habits to maintain stability.
Mild Symptom Detected	Mild distress detected but potential risk of escalation.	Suggest mindfulness guides, breathing exercises or lifestyle adjustment.
Significant Symptom Detected	High distress levels or long duration of symptoms.	Directs the user to medical help to seek a clinical consultation.

4.0 Monitoring Page

To monitor and track the user engagement and emotional stability over the time, the monitoring page will feature a continuous loop using the system clock and NLP analyzer as a sensor. Besides, each different user will have a different monitoring page to view their past engagement which is determined by how active the user is with using the app.

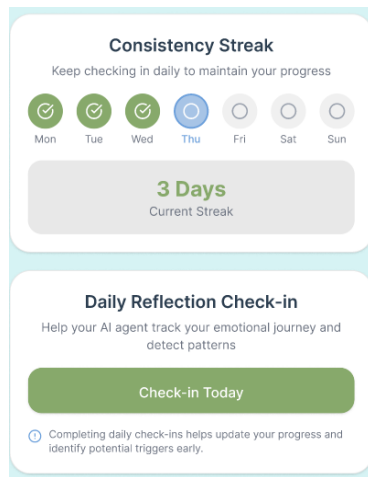


Figure 10: Daily check-in progress

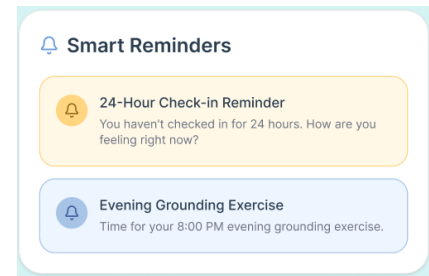
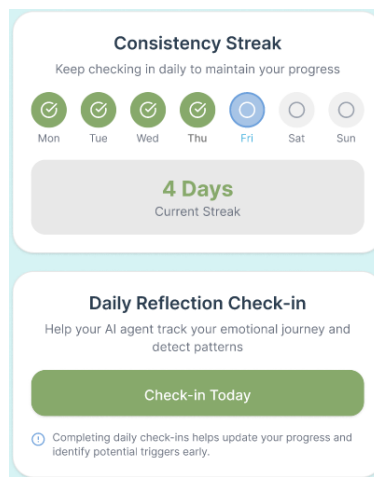


Figure 11: Smart reminder

To determine the activity status of a user, the system will track the daily check-in via a consistency streak based on the weekly progress bar. For every check-in, the system will actively update the user progress and change its status from “Unchecked” to “Active”. For every incomplete task, it will trigger a reminder to remind the user to complete their task no matter a daily check-in or daily exercise.

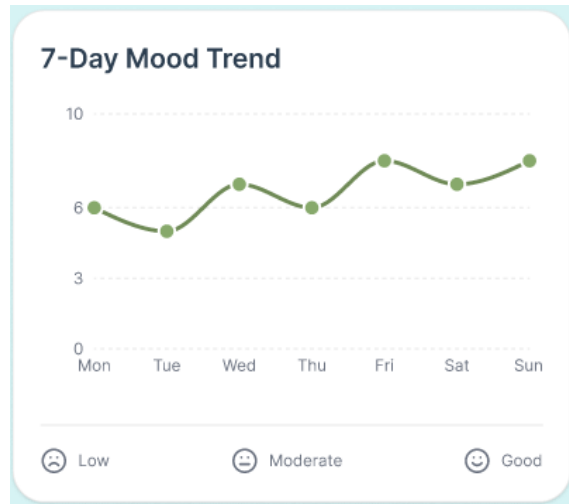


Figure 12: Mood trend graph

For every past activity, the system will translate raw NLP data into a useful insight. By analyzing the tone and frequency of chatbot interactions, the system generates a Mood Trend Graph which works the same as a mood tracker.

5.0 User Profile Page

The User Profile feature demonstrates how MindConnect can support personalization and private self-reflection in a single-user mental wellness application. The feature focuses on two core capabilities which are profile configuration and wellbeing history management. Through this feature, users are allowed to view and edit personal information such as name, age, and date of birth. It also enables the user to customize their preferences in the application such as the notifications setting and application theme. The users could also manage data options, including backup, export, and profile reset. Figure 13 and 14 shows the user view of the user profile page and what they can do on their personal information and preferences.

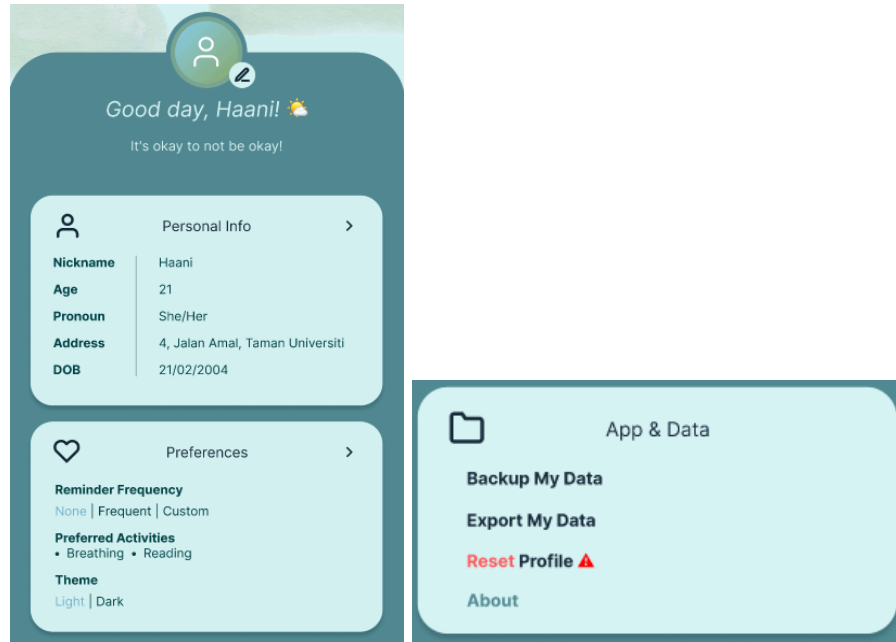


Figure 13: Personal, preferences, and data options on user profile page

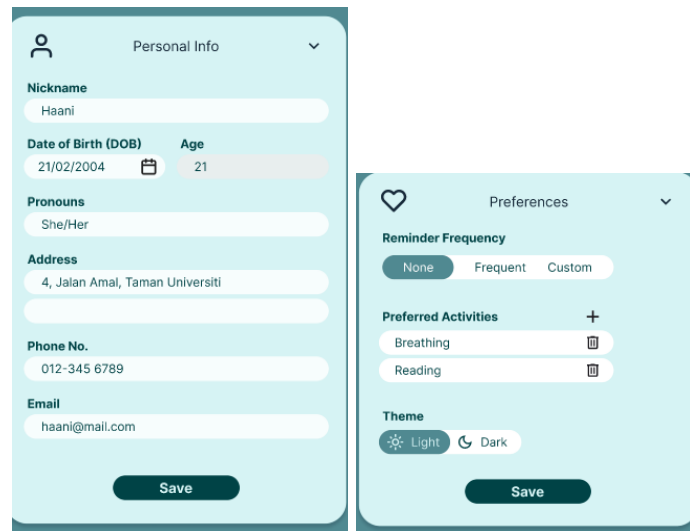


Figure 14: Personal and preferences information editor on user profile page

In addition to personal and preference information, users can also view their wellbeing history through the user profile feature. This allows them to monitor their past records and track improvements in their overall wellbeing over time. Figure 15 displays the wellbeing records with date and details.

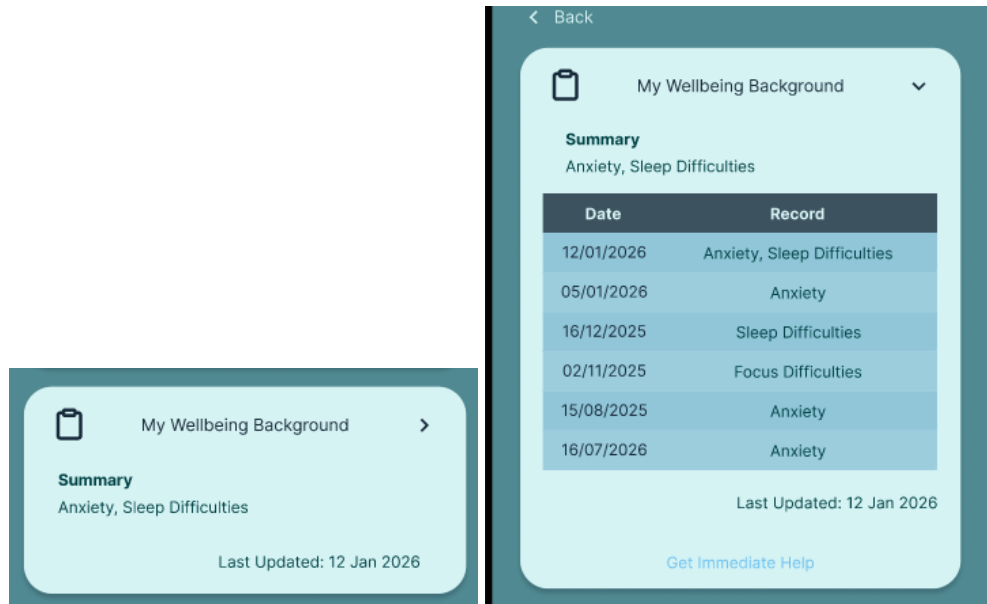


Figure 15: Users wellbeing record on user profile page

Conclusion

The development of MindConnect illustrates the successful integration of ethical Artificial Intelligence (AI) principles with practical implementation to create a useful mental health support system. Started by using First-Order Logic (FOL) to create the system's workflow, the system is capable of handling complex tasks such as preliminary diagnosis, therapy selection, and critical identification of crisis triage. Furthermore, the implementation of PEAS (Performance, Environment, Actuators, Sensors) model and State Space Search diagrams gives clear guidelines for how the AI will handle the user input and navigate towards appropriate output such as accurate emotion detection.

In addition, the implementation of specific logic for crisis triage shows that the system is built to prioritize the user's safety and health where high-risk input will trigger appropriate and logical responses. Last but not least, the Proof of Concept shows that the backend logic of the system can be transformed effectively into a user interface, featuring essential components such as an interactive AI Chatbox, progress monitoring page, and comprehensive questionnaire and user profile page.

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